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Exporting Liquidity: Branch Banking and Financial Integration

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ABSTRACT

Using exogenous liquidity windfalls from oil and natural gas shale discoveries, we demonstrate that bank branch networks help integrate U.S. lending markets. Banks exposed to shale booms enjoy liquidity inflows, which increase their capacity to originate and hold new loans. Exposed banks increase mortgage lending in nonboom counties, but only where they have branches and only for hard-to-securitize mortgages. Our findings suggest that contracting frictions limit the ability of arm's length finance to integrate credit markets fully. Branch networks continue to play an important role in financial integration, despite the development of securitization markets.

OVER THE PAST 30 YEARS, the banking system in the United States has gone through a significant transformation, relying more on capital markets and direct finance in funding loans and less on local bank deposits. The U.S. mortgage market has been at the forefront of this transformation, with 52% of loans in 2011 financed by securitization markets, up from 12% in 1980.[1] Moreover, improvements in information technology have facilitated bank lending well outside of branch-based geographical domains (Petersen and Rajan (2002)). These changes have allowed capital to flow across the U.S. economy and thus integrate local credit markets. The greater role of external capital markets in facilitating access to credit should have diminished the value of bank branch networks for lending. Yet over the same period the extent and density of bank

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[1] These statistics refer to the whole mortgage market, including mortgages for home purchase, home equity lines, as well as mortgage refinancings.

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offices and branches has continued to grow, from 63,200 (about five per bank) in 1990 to 89,800 (about 14 per bank) in 2012.[2]

In this paper, we show that branch networks still play an important role in integrating local credit markets. Using a unique positive exogenous shock to bank liquidity stemming from oil and gas "fracking" booms, we document that bank liquidity inflows increase mortgage lending in areas not experiencing the booms. Lending increases *only* in markets where banks have a branch presence. The effect is most pronounced for loan types that are subject to more contracting frictions, and therefore are harder to fund from external markets (e.g., through securitization). Moreover, banks exposed to shale booms seem to expand overall lending, as opposed to merely taking lending business away from other banks operating in similar markets. Combined, the results provide evidence that branch networks allow lenders to mitigate contracting frictions, and thus play an important role in integrating information intensive segments of credit markets where arm's length financing is limited.

To identify how funds flow across markets, we exploit the unexpected technological breakthrough that made vast amounts of shale oil and natural gas economically profitable to develop. Oil and gas companies pay significant mineral royalty payments to landowners to develop shale resources. This wealth windfall results in increased deposit supply for banks with branches in shale-boom counties (Gilje (2011) and Plosser (2011)). It also allows local landowners to pay down outstanding debt, further amplifying banks' liquidity windfalls.

Armed with this exogenous shock, we evaluate whether banks export liquidity by focusing on mortgage originations *outside* of shale-boom counties. Exploring mortgage lending has three advantages. First, these loans have a clear geographical dimension pinned down by the property location, which is not possible for other types of loans. Second, studying lending outside shale-boom counties alleviates concerns that shale discoveries drive credit demand. Third, the rich data set allows us to saturate models with county*year fixed effects, thus removing confounding demand effects. Conceptually, our analysis compares mortgage growth rates in the same county-year for two otherwise similar banks, one with branches in a shale-boom county (and thus exposed to a positive liquidity shock) and the other without such exposure.

Why might local liquidity shocks affect credit? The traditional banking literature argues that two frictions are necessary for a liquidity shock to propagate. Banks must face (liability-side) frictions in accessing external financing that preclude them from undertaking all profitable investment opportunities. In our setting, this friction stems from small and regional banks having limited access to external debt and equity markets and drawing most of their funds from insured (and potentially subsidized) deposits. These regional banks are at the center of our analysis, as they benefit most from the fracking wealth windfalls.[3] We show that shale booms lead to a simultaneous increase in the

[2] See http://www2.fdic.gov/hsob/HSOBRpt.asp.

[3] Our experiment is not likely to matter for very large banks, in part because such banks have relatively easy access to the interbank market, meaning that the marginal cost of funds is unlikely

quantity and decline in the cost of deposits at banks with branches in the shale-boom counties. The lower financing costs thus allow them to expand lending.

For the funding shock to stimulate new lending (that is, lending that other banks would not have originated), a second (asset-side) friction is required. Specifically, we need banks to have access to a set of borrowers over which they have a cost advantage relative to competing banks. A bank's physical branch footprint provides such an advantage, as geographical proximity lowers information asymmetries and allows better loan monitoring (e.g., Berger et al. (2005)). Thus, banks experiencing funding inflows should expand new credit in markets where they possess local information or a borrower relationship advantage. If distance did not matter, additional funding would affect lending in all nonshale counties equally. In contrast, if there are information asymmetries between banks and borrowers, and if these asymmetries are reduced by branches, then lending should increase more in counties where the banks have branches.

Our results indicate that banks export newly found liquidity to other markets. Furthermore, consistent with the notion that bank branches mitigate contracting frictions, mortgage lending increases *only* in outlying (nonboom) counties where exposed banks have branches. Lending does not increase in areas where exposed banks have no local knowledge and thus have no informational advantage over other sources of financing (e.g., securitization). The magnitudes are substantial. An average bank exposed to the boom grows mortgages in markets with branches 13% faster than a similar nonexposed bank, relative to the average mortgage growth rate of 11%. To further solidify this notion, we document that liquidity windfalls expand lending more in loan types subject to greater contracting frictions, which are less likely to be securitized, such as home equity lines (sold or securitized 4.5% of the time) and home purchase mortgages (sold or securitized 46% of the time), as opposed to mortgage refinancings (sold or securitized 65% of the time). Overall, our evidence suggests that bank branches integrate segments of lending markets that arm's length finance cannot.

Is this newly found liquidity allocated efficiently? Perhaps exposed banks waste the proceeds of the shale booms on pet projects net present value (NPV), as in Jensen (1986). One might even argue that such pet projects are most likely to be located near a bank's branches. We find no evidence supporting this agency problem interpretation. Loan delinquencies and charge-offs of banks exposed to shale booms fall rather than rise after exposure to the boom, with varying degrees of statistical significance that depend on the ex-post horizon and model specification.

Our findings contribute to several strands of the literature. First, the results extend research on the financial integration of U.S. markets and help explain why large benefits followed deregulation.[4] Two mechanisms, potentially

to be affected by a small shock to the deposit base. In fact, when we exclude the largest banks from our tests, the coefficients of interest do not change.

[4] The intrastate branching deregulation led to faster growth of the state economies (Jayaratne and Strahan (1996)) and lower growth volatility (Morgan, Rime, and Strahan (2004)). Such dereg-

working in parallel, can explain why the removal of restrictions on banks' ability to expand across geographical markets improved economic outcomes: tougher competition and improved capital mobility. Abundant evidence shows that increases in competition post-deregulation led to more efficient banking (Stiroh and Strahan (2003)), lowered the cost of capital for nonfinancial firms (Rice and Strahan (2010)), and contributed to better allocation of resources (Jayaratne and Strahan (1996)). There is much less direct evidence, however, about the effect of deregulation on capital mobility. In this paper, we show that branch networks contribute to capital flows across local credit markets. Thus, the increasing scope and density of bank branch networks made possible by deregulation potentially increased the efficiency of capital allocation by allowing savings in one area to finance investment in other areas.

Second, extant research evaluates whether close proximity between borrowers and lenders lowers the cost of information production and monitoring. Breakthroughs in information technology allowed for larger distances between borrowers and lenders (Petersen and Rajan (2002)). However, local lenders still extend more credit to riskier borrowers than distant lenders: loan rates tend to decline with the distance between borrower and lender (Degryse and Ongena (2005), Agrawal and Hauswald (2010)); more opaque (smaller) borrowers tend to establish enduring relationships with their local (small) banks; and larger, more transparent firms tend to borrow from larger (not so local) financial intermediaries (e.g., Berger et al. (2005)). In mortgage finance, locally concentrated lenders focus on soft information intensive segments of the mortgage market (Loutskina and Strahan (2011)) and have an advantage in screening and monitoring riskier borrowers (Ergungor (2010), Cortes (2011)). We contribute to this literature by documenting that even in the most developed, integrated, and technologically advanced lending market—the U.S. mortgage market—local finance is hard to substitute. Branch networks, and by extension local knowledge, remain important for segments of the credit markets subject to contracting frictions.

Third, our paper offers a microeconomic approach to testing the bank lending channel.[5] Most existing studies exploit a common bank liquidity shock engineered by a central bank. These shocks naturally correlate strongly with credit demand and business cycle conditions, creating identification challenges. Some studies address these challenges by exploiting cross-sectional differences in bank on-balance-sheet lending responses to aggregate liquidity shocks (e.g., Gertler and Gilchrist (1994), Kashyap et al. (1994), Kashyap and Stein (2000), Campello (2002), and Loutskina (2011)). Other more recent studies use natural

ulation resulted in better quality lending (Jayaratne and Strahan (1996)), more entrepreneurship and a greater share of small establishments (Black and Strahan (2002), Cetorelli and Strahan (2006), Kerr and Nanda (2009)), lower income inequality, less labor market discrimination, and weaker labor unions (Black and Strahan (2001), Beck, (2010), Levine and Levkov). That said, Loutskina and Strahan (2015) provide evidence from the recent housing boom that financial integration helped fuel local housing and economic booms, thus raising local volatility.

[5] See the theoretical arguments in Bernanke and Blinder (1988), Holmstrom and Tirole (1997), and Stein (1998).

experiments, where external shocks from abroad propagate into domestic credit markets through cross-border ownership of banks (e.g., Peek and Rosengren (1997), Schnabl (2012), Cetorelli and Goldberg (2012)). Our study is closest to those evaluating how local liquidity shocks from bank failures, government interventions, or bank runs affect lending supply (Ashcraft (2006), Khwaja and Mian (2008), Paravisini (2008), Iyer and Peydro (2011)). Unlike much of the earlier literature, however, we isolate the supply effects by exploiting data with precise information on the location of both lender and borrower.

In the remainder of the paper, Section I describes briefly the shale booms and their effects on local banks. Section II describes our data. Section III discusses our empirical methods and reports the results. Section IV contains a brief conclusion.

I. Shale Booms

In 2003, a surprise technological breakthrough combined horizontal drilling with hydraulic fracturing ("fracking"), enabling development of natural gas shale. The subsequent development of shale led to a new energy resource equivalent to 42 years of U.S. motor gasoline consumption. As recently as the late 1990s, shale gas was not thought to be economically viable, and represented less than 1% of U.S. natural gas production. The development of the Barnett Shale near Fort Worth, TX in 2003 changed industry notions about the viability of natural gas shale.

The Barnett Shale was initially drilled by Mitchell Energy in the early 1980s (Yergin (2011)). Rather than encountering the highly porous rock of a conventional formation, however, Mitchell encountered natural gas shale. While shale holds vast amounts of natural gas, it is highly nonporous and traps the gas in the rock. After 20 years of experimentation, in the early 2000s Mitchell Energy found that fracking could break apart shale and free natural gas for collection at the surface. This breakthrough combined with horizontal drilling and higher natural gas prices made large new reserves from shale economically profitable to develop.

The size of this energy resource and the low risk of unproductive wells ("dry holes") have led to a land grab for mineral leases. Before commencing any drilling operations, oil and gas firms must negotiate leases with mineral owners. Typically these contracts consist of a large upfront bonus, paid whether the well is productive or not, plus a royalty percentage based on the value of the gas produced over time. The resulting wealth windfalls led to large increases in local bank deposits. In an interview with the *Houston Chronicle*, H.B. "Trip" Ruckman III, president of a bank in the Eagle Ford Shale, stated that "We have had depositors come in with more than a million dollars at a whack." [6] This statement is consistent with reports of leasing terms. For example, an individual who owns one square mile of land (640 acres) and leases out his minerals at

[6] See *Houston Chronicle*, "Eagle Ford Banks Challenged as Deposits Skyrocket," June 8, 2012.

$10,000/acre would receive an upfront one-time payment of $6.4 million plus a monthly payment equal to 25% of the value of all the gas produced on his lease.

Two previous studies test how the shale booms affected banks and bank lending, as well local real outcomes. Plosser (2011) studies the impact on banks, finding that exposure to shale booms comes with increased bank lending as well as holdings of securities. These results are consistent with the idea that bank financing costs decline with the advent of shale booms. Gilje (2011) studies outcomes for nonfinancial firms within shale-boom counties themselves, finding that financially dependent industries grow relative to less dependent ones; he argues that greater credit supply within the booming areas stimulated investment. Neither study, however, explores the implications of the shale booms for outlying markets connected via branch networks, as we do.

The shale-boom windfalls represent an exogenous liquidity shock relative to the underlying characteristics of the affected communities for a number of reasons. First, the economic viability of shale wells was determined by larger macroeconomic forces, such as demand for natural gas and natural gas prices (Lake et al. (2013)), and therefore was unrelated to local economic conditions (health, education, demographics, etc.).

Second, the technological breakthroughs, namely horizontal drilling and hydraulic fracturing, were unexpected, and the viability of these technologies in different geographies was uncertain. It was extremely challenging even for oil and gas companies to predict how many wells an area might need to develop recoverable resources. Highlighting the fast pace and unpredictable nature of these discoveries, in 2008, *five* years after the technology was discovered, the Haynesville Shale area in Louisiana experienced an increase in lease bonus payments from a few hundred dollars an acre to $10,000 to $30,000 an acre within a one-year time period.[7] Combined, these facts suggest that it was unlikely that banks could strategically alter branch structures to gain greater exposure to shale liquidity windfalls. Thus, bank windfalls from shale discoveries are an attractive setting to study how liquidity is exported across branch networks in the United States.

II. Data and Sample Selection

Our sample is based on lending activity in the seven states with major shale discoveries between 2003 and 2010: Arkansas, Louisiana, North Dakota, Oklahoma, Pennsylvania, Texas, and West Virginia. As Figure 1 shows, each state contains a large number of counties that experienced shale booms as well as a large number of nonboom counties. Across the seven states, 124 counties experienced booms and 515 did not. Our sample, built at the bank-county-year level, includes all banks making housing-related loans (home purchase mortgages, mortgages for refinancing, and home equity loans) in any of these seven states.

[7] See the *Times-Picayune*, "SWEET SPOT; A Recent Rush on Natural Gas Drilling in Northwest Louisiana Is Turning Many Landowners into Instant Millionaires, and Stoking Others' Hopes," September 17, 2008.

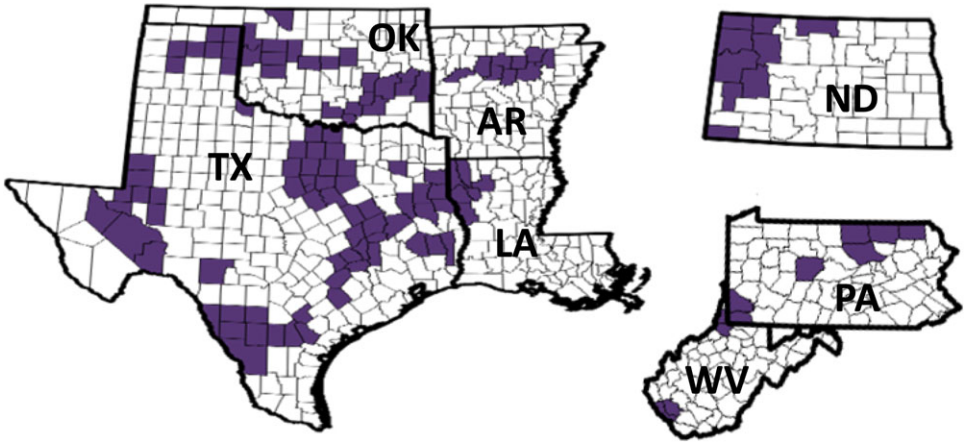

Figure 1. Location of shale activity. The figure maps the counties of the seven shale-boom states included in this study: AR, LA, ND, OK, PA, TX, and WV. White counties are nonboom counties while shaded counties are shale-boom counties as of 2010.

We consider all lenders irrespective of their branch locations (i.e., including loans originated without brick and mortar presence in a county) or exposure to the booms. We drop all nonbank lenders because most fund mortgage lending with securitization and thus are only affected by changes in the aggregate supply of funds from the securitization market. The sample begins in 2000 (three years before the first shale boom) and ends in 2010.

Using the Summary of Deposits from the Federal Insurance Deposit Corporation (FDIC), we determine the number of branches and amount of deposits held by each bank in each county-year in the seven states.[8] These data allow us to measure the shale-boom shock using *Share of Branches in Boom Counties*, which is equal to the fraction of branches owned by each bank that are located in a shale-boom county. The measure ranges from zero (for banks without branches in boom counties, or for banks with branches in boom counties during the years prior to a boom's onset) to one (for banks with all of their branches in boom counties after the onset of the booms). This variable equals zero for all bank-years prior to 2003, the year of the first shale investment. After 2003, the variable increases within bank over time as more counties experience booms.

Our models focus on the effect of exposure to the shale boom on mortgage credit growth, but we include other bank characteristics as control variables, each measured from the end of the prior year. These variables include: *Log of Assets*$_{t-1}$, *Deposits*/*Assets*$_{t-1}$, *Cost of Deposits*$_{t-1}$ (= interest expenses on deposits/total deposits), *Liquid Assets*/*Assets*$_{t-1}$, *Capital*/*Assets*$_{t-1}$ (= Tier 1 capital/ assets), *C&I Loans*/*Asset*$_{t-1}$, *Mortgage Loans*/*Assets*$_{t-1}$, *Net Income*/*Assets*$_{t-1}$, *Loan Commitments*/*Assets*$_{t-1}$, and *Letters of Credits* /*Assets*$_{t-1}$. Data for bank control variables come from year-end Call Reports.

[8] See http://www2.fdic.gov/sod/.

Table I

Summary Statistics

This table reports summary statistics for banks operating in states with counties exposed to the shale boom. Observations are at the bank-year level in Panels A to C, and at the bank-county-year level in Panel D. The sample comes from the seven states that experienced shale booms between 2000 and 2010. *Share of Branches in Boom Counties* equals the fraction of a bank's branches located in shale-boom counties (set to zero before the onset of a shale boom). The distribution of bank branches comes from the FDIC Summary of Deposits, which we use to determine whether a branch (or the bank that owns it) is exposed to the boom. Bank characteristics come from year-end Call Reports. Growth in mortgage originations comes from annual HMDA data.

	Nonexposed Banks		Exposed Banks	
	Mean	*SD*	Mean	*SD*
Panel A: Exposure to Deposit Shock (Bank-Year Variation)				
Share of branches in boom counties	0	0	0.45	0.39
Number of bank-years	7,451		1,280	
Number of banks	1,350		357	
Panel B: Bank Characteristics (Bank-Year Variation)				
Deposit growth	0.085	0.141	0.102	0.156
Log of assets	12.451	1.390	13.428	2.090
Deposits/assets	0.827	0.086	0.827	0.087
Cost of deposits	0.022	0.010	0.017	0.008
Liquid assets/assets	0.273	0.148	0.225	0.132
Capital/assets	0.099	0.028	0.101	0.029
C&I loans/asset	0.112	0.090	0.145	0.087
Mortgage loans/assets	0.347	0.137	0.323	0.117
Net income/assets	0.009	0.008	0.008	0.011
Loan commitments/assets	0.109	0.140	0.134	0.182
Letters of credits/assets	0.006	0.011	0.010	0.020
Panel C: Annual Mortgage Growth Rates (Bank-Year Variation)				
Growth in mortgage originations	0.112	0.621	0.117	0.576
Growth in retained mortgages	0.077	0.672	0.091	0.618
Growth in sold mortgages	0.107	1.159	0.098	1.097
Panel D: Annual Mortgage Growth Rates (Bank-County-Year Variation)				
Growth in mortgage originations	0.023	1.123	0.027	1.058
Growth in retained mortgages	0.013	1.200	0.017	1.159
Growth in sold mortgages	0.039	1.117	0.036	1.014

Table I reports summary statistics for our measure of banks' exposure to the shale well boom—*Share of Branches in Boom Counties* (Panel A), as well as the lagged bank characteristics (Panel B), separated by whether the bank has any exposure to a shale-boom county. The table shows that exposed banks tend to be larger than nonexposed banks and that their deposits grow faster and have lower cost, consistent with the notion that exposure to the shale boom leads to increases in deposit supply. The marked difference in asset size (log of assets)

is a potential concern in our models because large banks differ in many ways from smaller ones; in robustness tests, we filter out larger banks.

To measure mortgage activity, we use the detailed data on mortgage applications collected annually under the Home Mortgage Disclosure Act (HMDA), which we merge with Call Report data as in Loutskina and Strahan (2009). Whether a lender is covered depends on its size, the extent of its activity in a Central Business Statistical Area (CBSA), and the weight of residential mortgage lending in its portfolio.[9] The HMDA data include loan size, whether a loan was approved, as well as some information on borrower characteristics. Using HMDA data, we measure mortgage origination growth by bank-county-year. HMDA reports both the identity of the lender as well as the location of the property down to the census-tract level. These are the only comprehensive data on lending by U.S. banks that allow researchers to locate borrowers geographically. In principle we would also like to test for similar effects on other kinds of loans (especially loans to small businesses), but microdata at the loan level are not available outside of housing. HMDA also contains information on the purpose of the loan (mortgage purchase loans, home equity loans, and mortgage refinancings) and whether the lender expects to sell or securitize the loan within one year of origination. We use these data to test whether loans easier to finance in securitization markets respond less to the local liquidity inflows that follow shale booms.

Table I reports summary statistics for mortgage growth rates, defined as the difference in the log of mortgage originations, at both bank-year (Panel C) and bank-county-year (Panel D) levels. For purposes of simple comparisons, we focus mainly on the summary statistics at the bank-year level, since our main variables of interest vary *only* at that level; in our regressions, we absorb variation across county-years with fixed effects. For the average exposed bank, mortgages grow 11.7% per year, compared to 11.2% for nonexposed banks. This difference is larger for retained mortgage growth, which averages 9.1% per year for exposed banks, compared to 7.7% for nonexposed banks. These raw differences could be attributed to the deposit windfalls as well as to economic growth in the boom counties. We separate these two effects in our regressions. Note that the standard deviation (SD) of the mortgage growth rates is very high relative to the mean, but much of this variation reflects time-series fluctuations stemming from changes in interest rates (which alter refinancing rates drastically) as well as variation around the housing boom (2004 to 2006) and bust (2006 to 2010) periods, which our data straddle.

[9] Any depository institution with a home office or branch in a CBSA must report HMDA data if it has made a home purchase loan on a one- to four- unit dwelling or has refinanced a home purchase loan and if it has assets above $30 million. Any nondepository institution with at least 10% of its loan portfolio composed of home purchase loans must also report HMDA data if it has assets exceeding $10 million. Consequently, HMDA data does not capture lending activity of small or rural originators. U.S. Census data show that about 83% of the population lived in metropolitan areas over our sample period and hence the bulk of residential mortgage lending activity is likely to be reported under the HMDA.

HMDA also offers some borrower characteristics, which we use to build the following control variables for all loans originated at the bank-county-year level: borrower and area income, loan size-to-borrower-income ratio, percent women applicants, percent minority applicants, and percent minority population in the area of loan applications. In all of our models we control for the contemporaneous means of each of these borrower attributes across all loan applications in a given bank-county-year.

III. Methods and Results

A. Shale Booms as a Positive Liquidity Shock

We first establish that banks exposed to shale booms experience liquidity inflows. Such inflows occur both because local mineral rights owners expand the local supply of deposits by putting funds into local bank branches (which is directly observable) and because they pay back outstanding loans (which is not).

To establish the first channel—that bank deposit supply increases with shale-boom exposure—we report regressions of both deposit quantity (deposit growth) and price (interest expense on deposits/deposits), as follows:

$$\begin{aligned} \textit{Deposit Growth}_{i,t} = \gamma_1\, &\textit{Bank Boom Exposure}_{i,t} \\ &+ \textit{Control Variables} + \varepsilon_{i,t.}, \end{aligned} \tag{1a}$$

$$\begin{aligned} \textit{Interest Expense/Deposits}_{i,t} = \gamma_2 &\textit{Bank Boom Exposure}_{i,t} \\ &+ \textit{Control Variables} + \varepsilon_{i,t}, \end{aligned} \tag{1b}$$

where the unit of analysis varies by bank i and year t. If shale booms increase deposit supply, then we expect $\gamma_1 > 0$ and $\gamma_2 < 0$. We include lags of bank characteristics as control variables, as well as bank and year fixed effects. Standard errors are clustered by bank.

As shown in Table II, deposit quantity increases and its price falls with bank shale exposure, consistent with a positive supply shock. To understand magnitudes, consider a bank with average exposure to shale (*Share of Branches in Boom Counties* = 0.45) and one with no exposure. Compared to unexposed banks, exposed banks' deposits would grow about 2.5 percentage points faster (column (1): 0.45*0.0567) and interest expense on deposits would be about 7 basis points lower (column (2): 0.45*−0.0015). These magnitudes line up well with the differences in means between exposed and nonexposed banks in Table I.

Table II also reports similar regressions for the bank capital-to-asset asset ratio and the return on assets (ROA = Net Income/lagged Total Assets). These results suggest that neither bank equity capital nor ROA differ between exposed versus unexposed banks ex ante. Thus, other than bank size, the two sets of banks appear comparable along key observable characteristics.

Table II

Shale-Boom Exposure Expands Deposit Supply

This table reports bank-year regressions of the growth in deposits and the price of deposits (interest expense on deposits/deposits) on measures of exposure to shale-boom counties. Control variables are compiled based on Call Report data and are measured as of the prior year-end. Standard errors are clustered by bank. All regressions include bank and year fixed effects. *t*-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Deposit Growth	Price of Deposits(Interest Expense/Deposits)	Capital/ Assets	Net Income/ Assets
	(1)	(2)	(3)	(4)
Share of branches in boom counties$_t$	0.0567***	–0.0015***	–0.00204	0.000479
	(4.03)	(2.66)	(1.08)	(0.61)
Log of assets$_{t-1}$	–0.301***	0.0036***	–0.0193***	0.0039***
	(14.56)	(10.13)	(6.54)	(7.13)
Deposits/assets$_{t-1}$	–1.098***	0.0083***	–0.185***	0.003
	(13.58)	(4.43)	(8.65)	(0.89)
Liquid assets/assets$_{t-1}$	–0.0674	–0.0027**	–0.0163*	–0.0004
	(1.38)	(2.21)	(1.72)	(0.20)
C&I loans/asset$_{t-1}$	0.214***	–0.0003	–0.0251***	0.0065**
	(3.56)	(0.21)	(2.81)	(2.50)
Mortgages/asset$_{t-1}$	–0.002	0.0034**	–0.0521***	0.0128***
	(0.03)	(2.11)	(5.01)	(4.87)
Loan commitments/assets$_{t-1}$	0.0667	0.0006	0.0117	0.0043*
	(0.98)	(0.76)	(1.13)	(1.95)
Letters of credits/assets$_{t-1}$	0.103	–0.0101	0.0395	0.0192
	(0.23)	(1.42)	(0.73)	(1.54)
Bank fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	13,698	13,867	13,867	13,867
R^2	54.8%	89.5%	76.3%	63.4%

B. Baseline Result

To evaluate how liquidity windfalls affect mortgage lending, we estimate a three-dimensional panel regression of the growth in mortgage originations in nonshale counties on each bank's shale-boom exposure, as follows:

$$\begin{aligned} \textit{Mortgage Growth}_{i,j,t} = \alpha_{j,t} + \beta\ \textit{Share of Branches in Boom Counties}_{i,t} \\ + \textit{Borrower \& Lender Controls} + \varepsilon_{i,j,t}, \end{aligned} \tag{2}$$

where i indexes lenders, j indexes counties, and t indexes years. With this panel structure, we can absorb county*year effects ($\alpha_{j,t}$), thus removing time-varying, county-level demand-side shocks related to business cycles, industry composition, housing demand, etc. To further distinguish supply shocks from potentially confounding demand shocks, we include in our sample only counties that *did not* experience a shale boom during the 2000 to 2010 period. Unlike *Mortgage Growth*, *Share of Branches in Boom Counties* does not vary

across counties for a given bank-year; hence, we construct standard errors by clustering by bank throughout all of our analysis.

One potential source of bias that might not be captured by the county*year ($\alpha_{j,t}$) effects could stem from bank decisions to enter shale-boom markets. For example, after observing the advent of shale discoveries in 2003, banks might enter shale-boom counties (or counties with known shale reserves) to raise low-cost deposits. If such entry were motivated by the need to fund new loans in connected markets, then our effects could be driven by *both* supply and demand factors, thus invalidating our identification strategy. We allay this concern by estimating instrumental variable (IV) models in which we use *Share of 2002 Branches in Boom Counties*$_{i,t}$—a bank's exposure that would have occurred based on its preboom 2002 branch network—as an instrument for the actual *Share of Branches in Boom Counties*$_{i,t}$. We show that the instrument is powerful, easily passing tests for weak instruments. It meets the exclusion restriction because banks could not plausibly have anticipated shale booms in 2002, since not even experts in the energy sector predicted its advent before 2003.

We use growth in mortgage origination as the dependent variable, rather than the log level of originations (or market share).[10] The problem with the latter approach is that models based on levels (even with bank*county fixed effects) "difference out" average behavior for a bank-county *across the whole sample*, whereas the growth rate approach differences out lending for a bank-county *from the prior year*. This matters because the mortgage data exhibit very high origination levels early in the sample (2000 to 2007), followed by a crash later in the sample (2008 to 2010). Estimating the model in growth rates helps alleviate this problem, since bank-specific "trends" get differenced out in the growth rates. Also, by normalizing the dependent variable by last year's lending in a respective bank-county, we accommodate significant within-bank variation in the extent of lending in different counties.[11]

We next decompose *Mortgage Growth* into the growth in retained mortgages and the growth in sold or securitized mortgages. Contracting frictions make some loan securitization difficult, however, either because lenders have better information than investors or because incentives for lenders to engage in sufficient monitoring would diminish if a loan were sold (e.g., Gorton and Pennacchi (1995), Holmstrom and Tirole (1997), Keys et al. (2010)). Increases in liquidity thus should increase mortgages that must be held by the originating lender, as opposed to those easily sold to arm's length investors. Analysis of retained versus sold loans allows us to evaluate whether banks merely retain more loans

[10] Results in levels are available in the Internet Appendix, which may be found in the online version of this article.

[11] In addition to the intensive margin, we explore the variation on the extensive margin, which our growth rate approach cannot address since we require positive lending in the prior year (not reported). We do not find that liquidity windfalls affect the extensive margin. We believe that this null result occurs because the liquidity shocks matter only in areas where banks have a branch presence (documented below), and banks almost always lend a nonzero amount in areas where they have branches. We have also tested whether mortgage acceptance rates vary with shale-boom exposure and find that they do not.

in response liquidity inflows at the expense of the secondary market. Furthermore, this decomposition allows us to document the true on-balance-sheet sensitivity of total lending to a liquidity supply shock.

Table III reports the ordinary least squares (OLS) and IV results of estimating equation (2), along with the first-stage regression. We find significant positive effects of exposure to shale booms on both total mortgage growth (columns (2) and (3)) and growth of retained mortgages (columns (5) and (6)), but no effect on sold-loan growth (columns (8) and (9)). In all three cases we have very strong identification, as bank branch networks are quite stable over time. The t-statistic on the instrument for *Share of Branches in Boom Counties* in the first-stage regressions exceeds 23 in all three cases. We also estimate similar signs and statistical significance between OLS and IV, although the OLS coefficients have larger magnitudes. For example, the OLS estimate of the effect of *Share of Branches in Boom Counties* on total mortgage origination growth equals 0.146, compared to 0.068 using IV (compare columns (2) and (3)). Below we show that the differences between the OLS and IV estimates disappear entirely when we divide the data based on bank branch presence.

C. Local versus Nonlocal Markets

Our core evidence indicates that contracting frictions affect the flow of funds in the mortgage market. Extant literature suggests that local lenders have an informational advantage as they tend to lend to more opaque and riskier firms (Berger, et al. (2005)). Mortgage lenders with branches near their borrowers also have an advantage in monitoring borrowers that may experience distress. Following these arguments, we next evaluate whether liquidity windfalls increase mortgage originations more in counties where banks have branches, as compared to counties where they lend without a brick and mortar presence.[12]

In Table IV, we separate our sample into markets where banks have branches (local) versus those where they do not (nonlocal). In local markets, the coefficient of interest equals 0.288 in OLS versus 0.313 in IV.[13] In nonlocal markets, we find *no* evidence that exposure to shale booms affects mortgage originations. Mortgage growth increases for banks exposed to shale-boom windfalls, but *only* for local banks—those with branches in the same county as the property being financed. The estimate suggests that a typical exposed bank (e.g., one with about 45% of its branches in a shale-boom county—recall Table I) would grow its mortgage portfolio in local markets 13 percentage points (= 0.45*0.288) faster than a similar bank without exposure to the shale-boom windfalls (based on the coefficient of interest in column (1) of Table IV). Table IV thus establishes

[12] A natural way to sort our data would be based on whether or not Fannie or Freddie provide credit guarantees, such as comparing jumbo and nonjumbo mortgages. Unfortunately, the markets we study have low real estate prices so that the vast majority of loans fall into the nonjumbo category.

[13] The fact that the OLS and IV estimates are close suggests that shale boom exposure is not driven by credit demand. In an earlier draft, in fact, we showed that banks did not expand into shale boom areas in response to expansion of past lending.

Table III

Effect of Liquidity Supply on Mortgage Lending

This table reports regressions of the percentage change in mortgage originations by bank-county-year on measures of exposure to shale-boom counties. Bank-county-years are excluded if the county experienced a shale boom over the sample period. Regressions include both lender (reported) and borrower (not reported) control variables. Lender controls are from the Call Reports from the prior year; borrower controls are average borrower and area income, loan size-to-income ratio, percent women applicants, percent minority applicants, and percent minorities in the area for loans made during the current year (from HMDA). Standard errors are clustered by bank. All regressions also include county*year fixed effects. *t*-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Full Sample			Retained Loans			Sold Loans		
	First Stage	OLS	IV	First Stage	OLS	IV	First Stage	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Share of 2002 branches in boom counties$_t$	0.870***	–	–	0.838***	–	–	0.897***	–	–
	(36.79)	–	–	(36.91)	–	–	(23.73)	–	–
Share of branches in boom counties$_t$	–	0.146**	0.0684*	–	0.325**	0.147***	–	0.202	0.0585
	–	(2.17)	(1.73)	–	(2.26)	(2.61)	–	(1.26)	(1.07)
Log of assets$_{t-1}$	0.00366	–0.0121	–0.0120***	0.0034	–0.0096	–0.00927***	0.00483	–0.0217**	–0.0219***
	(1.60)	(1.45)	(5.69)	(1.62)	(1.11)	(3.44)	(1.58)	(2.10)	(7.61)
Deposits/assets$_{t-1}$	(0.03)	(0.07)	–0.0724**	(0.02)	(0.31)	–0.308***	(0.06)	0.06	0.05
	(0.78)	(0.41)	(1.99)	(0.53)	(1.40)	(6.03)	(1.01)	(0.31)	(1.20)
Cost of deposits$_{t-1}$	0.55	(1.27)	–1.287**	0.57	(0.88)	(0.87)	0.85	(6.300)	–6.382***
	(1.33)	(0.43)	(2.01)	(1.32)	(0.24)	(1.04)	(1.08)	(1.37)	(6.51)
Liquid assets/assets$_{t-1}$	0.0463	0.151	0.152***	0.0477	–0.0596	–0.0493	0.0379	0.497**	0.498***
	(0.83)	(1.08)	(3.43)	(1.01)	(0.31)	(0.85)	(0.54)	(2.58)	(8.10)
Capital/assets$_{t-1}$	–0.132	–1.442**	–1.462***	–0.117	–0.243	–0.299*	–0.262	–1.987***	–2.044***
	(1.06)	(2.20)	(11.70)	(0.96)	(0.39)	(1.84)	(1.43)	(2.92)	(11.73)
C&I loans/assets$_{t-1}$	0.0559	0.081	0.0867	0.0527*	–0.0453	–0.0237	0.0685	0.214	0.233***
	(1.51)	(0.54)	(1.43)	(1.66)	(0.16)	(0.28)	(1.05)	(0.70)	(2.63)
Mortgage loans/assets$_{t-1}$	0.136	0.0445	0.0506	0.106	–0.0104	0.0129	0.19	0.289*	0.308***
	(1.12)	(0.29)	(1.25)	(1.07)	(0.05)	(0.24)	(1.24)	(1.73)	(5.68)
Net income/assets$_{t-1}$	–0.0589	4.441**	4.482***	–0.0679	5.606*	5.676***	–0.321	2.474	2.503***
	(0.22)	(2.29)	(10.90)	(0.26)	(1.76)	(9.98)	(0.62)	(0.76)	(4.51)

(*Continued*)

Table III—*Continued*

	Full Sample			Retained Loans			Sold Loans		
	First Stage	OLS	IV	First Stage	OLS	IV	First Stage	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Loan commitments/assets$_{t-1}$	0.00285	0.0301	0.0300***	0.00265	–0.0191	–0.0176**	0.00484	0.0218	0.0203*
	(0.55)	(1.32)	(4.87)	(0.54)	(0.51)	(2.14)	(0.57)	(0.57)	(1.72)
Letters of credits/assets$_{t-1}$	–0.331**	0.787	0.742***	–0.278**	0.402	0.319	–0.454**	2.525**	2.434***
	(2.32)	(0.94)	(3.78)	(2.45)	(0.47)	(1.33)	(2.29)	(2.25)	(8.85)
Borrower controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County*year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank-clustered standard errors	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen-Paap test for weak instruments	–	–	59,757	–	–	44,600	–	–	28,163
Observations	92,144	92,144	92,144	71,034	71,034	71,034	49,427	49,427	49,427
R^2	74.5%	7.3%		74.8%	7.9%		71.2%	13.0%	

Table IV

Effect of Liquidity Supply on Mortgage Lending: Local versus Nonlocal Lenders

This table compares regressions of the percentage change in mortgage originations by bank-county-year on measures of exposure to shale-boom counties for local versus nonlocal markets. Local markets are those in which a bank has at least one branch. Bank-county-years are excluded if the county experienced a shale boom over the sample period. Regressions include both lender (reported) and borrower (not reported) control variables. Lender controls are from the Call Reports from the prior year; borrower controls are average borrower and area income, loan size-to-income ratio, percent women applicants, percent minority applicants, and percent minorities in the area for loans made during the current year (from HMDA). Standard errors are clustered by bank. All regressions also include county*year fixed effects. *t*-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Local		Nonlocal	
	OLS (1)	IV (2)	OLS (3)	IV (4)
Share of branches in boom counties$_t$	0.288***	0.313***	0.092	0.039
	(2.94)	(3.32)	(0.60)	(0.87)
Log of assets$_{t-1}$	0.009	0.00902**	–0.0161*	–0.0138***
	(1.08)	(2.42)	(1.68)	(4.99)
Deposits/assets$_{t-1}$	0.411***	0.411***	(−0.16)	–0.139***
	(3.97)	(5.07)	(0.81)	(3.10)
Cost of deposits$_{t-1}$	–1.481	–1.476	(−1.15)	1.386
	(0.27)	(1.17)	(0.36)	(1.57)
Liquid assets/assets$_{t-1}$	0.271**	0.271***	0.128	0.150***
	(2.25)	(3.60)	(0.74)	(2.63)
Capital/assets$_{t-1}$	–0.323	–0.323	–1.837***	–1.769***
	(0.50)	(1.52)	(2.59)	(11.03)
C&I loans/asset$_{t-1}$	0.200	0.198*	0.056	0.178**
	(1.25)	(1.81)	(0.32)	(2.33)
Mortgage loans/assets$_{t-1}$	0.064	0.065	0.0244	0.022
	(0.40)	(0.78)	(0.15)	(0.44)
Net income/assets$_{t-1}$	2.013	2.000***	4.718**	5.363***
	(0.79)	(2.61)	(2.22)	(10.06)
Loan commitments/assets$_{t-1}$	–0.106	–0.106**	0.0347	0.0305***
	(0.77)	(2.05)	(1.42)	(4.30)
Letters of credits/assets$_{t-1}$	–0.232	–0.220	1.035	0.901***
	(0.33)	(0.60)	(1.03)	(3.64)
Borrower controls	Yes	Yes	Yes	Yes
County*year fixed effects	Yes	Yes	Yes	Yes
Bank-clustered standard errors	Yes	Yes	Yes	Yes
Kleibergen-Paap test for weak instruments	–	12,241	–	40,987
Observations	22,192	22,192	70,002	70,002
R^2	20.2%		8.5%	

that local windfalls stimulate lending only in markets connected through bank branching networks.

D. Results by Mortgage Type

Next, we evaluate the effect of the windfalls by mortgage type. Contracting frictions should be most pronounced for home equity loans (because these are often subordinated), and least for mortgage refinancing (because borrowers have an established payment history), with home purchase originations between these two extremes. Consistent with this notion, securitization rates are lowest among home equity loans (4.5%) and highest among mortgage refinancing loans (65%) with mortgages for home purchase in the middle (46%). Hence, we expect the liquidity inflows to matter most for home equity loans and least for mortgage refinancings.

To test this idea, Table V splits our sample by both loan type and market type.[14] We report only the coefficients of interest, but the specifications include the same set of borrower and lender controls and county*year fixed effects as the previous sets of results. As before, we report results using both OLS and IV. Consistent with the earlier analysis, only local lenders respond to the liquidity windfalls. Moreover, their response is evident only among loans that are hard to securitize (and subject to more contracting frictions): mortgages for home purchase and home equity loans, but not mortgages for refinancing. In these specifications, the effects of the windfall are largest for the home equity segment, intermediate for mortgages for home purchase, and zero for the refinancing segment.[15]

E. Is New Mortgage Lending a Free Cash-Flow Agency Problem?

Our results suggest that portions of the mortgage market where arm's length finance is limited by information frictions respond positively to local liquidity shocks. This increase, however, could reflect lender agency problems (Jensen (1986)), whereby unexpected cash inflows lead managers to overinvest in value-destroying loans (i.e., marginal loans have NPV < 0). The agency explanation is hard to rule out because we are not able to follow loan outcomes at the

[14] Samples differ across the three columns in Table V because we model the growth rate in lending, so a bank-county only appears if there are nonzero originations in two consecutive years.

[15] We evaluate whether the effect of the liquidity windfalls shifts systematically during the 2008 financial crisis. We find no significant changes. This may seem surprising because securitization of subprime and jumbo mortgages was dramatically reduced by the crisis, suggesting that availability of local funds should have become more important postcrisis. But banks in our sample operate in areas with relatively low-cost housing where Fannie and Freddie dominate, as opposed to the high-priced markets on the coasts. The government sporsored enterprises also substantially increased their role in providing financial subsidies during and after the crisis. Moreover, the housing boom/bust cycle and expansion of subprime credit was much less pronounced in these states compared to regions like southern California or Florida.

Table V

Effect of Liquidity Supply on Mortgage Lending: Home Purchases, Refinancings, and Home Equity Loans

This table compares regressions of the percentage change in mortgage originations by bank-county-year on measures of exposure to shale-boom counties for the local versus nonlocal market. Local markets are those in which a bank has at least one branch. Bank-county-years are excluded if the county experienced a shale boom over the sample period. Regressions include both lender (reported) and borrower (not reported) control variables. Lender controls are from the Call Reports from the prior year; borrower controls are average borrower and area income, loan size-to-income ratio, percent women applicants, percent minority applicants, and percent minorities in the area for loans made during the current year (from HMDA). Standard errors are clustered by bank. All regressions also include county*year fixed effects. t-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Local Banks		Nonlocal Banks	
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
Panel A: Mortgages for Purchase				
Share of branches in boom counties$_t$	0.277***	0.219**	0.07	0.03
	(2.87)	(2.05)	(0.85)	(0.56)
Kleibergen-Paap test for weak instruments	–	11,616	–	24,496
Observations	20,755	20,755	44,126	44,126
R^2	19.8%		11.4%	
Panel B: Home Equity Loans				
Share of branches in boom counties$_t$	0.320***	0.294**	–0.24	–0.06
	(2.65)	(2.08)	(1.23)	(0.49)
Kleibergen-Paap test for weak instruments	–	10,497	–	9,008
Observations	18,519	18,519	16,339	16,339
R^2	24.9%		25.0%	
Panel C: Mortgages for Refinancing				
Share of branches in boom counties$_t$	0.09	0.0791	0.199*	0.04
	(0.83)	(0.67)	(1.80)	(0.63)
Kleibergen-Paap test for weak instruments	–	11,178	–	21,951
Observations	20,024	20,024	46,256	46,256
R^2	30.9%		15.6%	

bank-county-year level, thus precluding our preferred identification strategy.[16] We can, however, measure outcomes for retained loans at the overall bank-year level based on data from Call Reports. If agency problems are driving the increase in lending, then lenders should have higher loan losses after being exposed to shale booms.

[16] Loan level data on delinquencies and foreclosures are available, but inferring which investor actually bears losses is not. For example, when loans that have been securitized (or loans where originators have purchased credit protection from one of the GSEs) go bad, losses may not affect the originating lender, or such losses may be shared with other investors.

Table VI

Do Shale-Boom Exposed Banks Make Bad Loans?

This table reports bank-year regressions of mortgage charge-offs and delinquencies (loans 90+ days late) relative to total mortgages on measures of exposure to shale-boom counties. Regressions include both lender controls from the Call Reports from the prior year. Standard errors are clustered by bank. All regressions also include bank and year fixed effects. t-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Delinquencies$_{t+1}$/ Mortgages$_t$		Charge-Offs$_{t+1}$/ Mortgages$_t$	
	OLS	IV	OLS	IV
Share of branches in boom counties$_t$	–0.00798**	–0.0042***	–0.00324***	–0.0015***
	(2.46)	(2.99)	(3.81)	(3.07)
Log of assets$_{t-1}$	0.00974***	0.00356***	0.00285***	0.00142***
	(5.51)	(4.97)	(4.76)	(5.59)
Deposits/assets$_{t-1}$	0.01	–0.00972**	0.001	0.00352**
	(0.75)	(2.21)	(0.70)	(2.25)
Liquid assets/assets$_{t-1}$	–0.0747***	–0.00950***	–0.0187*	0.000601
	(2.83)	(2.68)	(1.66)	(0.48)
C&I loans/asset$_{t-1}$	–0.0440***	–0.00826*	–0.0125**	0.00123
	(2.61)	(1.66)	(2.27)	(0.69)
Mortgage loans/assets$_{t-1}$	–0.0798***	–0.0110***	–0.0244**	0.00211
	(3.39)	(2.70)	(2.05)	(1.45)
Loan commitments/assets$_{t-1}$	–0.0186**	–0.0111***	–0.00862*	–0.00487***
	(2.04)	(4.77)	(1.76)	(5.90)
Letters of credits/assets$_{t-1}$	–0.0254	–0.0383*	–0.0115	–0.00571
	(0.71)	(1.66)	(0.59)	(0.70)
Bank fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Kleibergen-Paap test for weak instruments		6,973		6,973
Observations	12,999	12,999	12,659	12,659
R^2	51.9%		36.1%	

Table VI reports regressions of the fraction of mortgage loans that were charged off or are delinquent (90 days or more past due or nonaccruing) in year $t + 1$ as a fraction of mortgage balances on bank balance sheets in year t, as a function of shale-boom exposure. We include the same set of lagged bank characteristics as in Table II, along with bank and year fixed effects. We also report both OLS and IV, as before. The results provide no support for the agency-based explanation. In fact, we find that loan performance is better at banks with exposure to the shale booms.

F. Robustness Tests

The baseline results withstand a wide set of robustness tests reported in the Internet Appendix. First, we evaluate whether our results could be attributed to the shale-boom exposed banks systematically lending more irrespective of the boom. That is, we test whether the parallel trends assumption between

"treatment" and "control" banks is violated by comparing whether banks exposed and not exposed to booms behave similarly before the booms actually occur. We create the variable *Preboom Indicator for Booming Banks* that equals one for booming banks during all years prior to an actual boom. For example, for a bank that first became exposed to a shale boom in 2007, the indicator would be set to one during 2000 to 2006. The indicator would be equal to zero for all years after 2006 in this example. For banks that never experience exposure, the indicator equals zero for all years. By introducing this variable, we can rule out the possibility that banks that experience booms (the treatment group) behave differently from other banks (control group) during "normal" times. Consistent with this notion, the coefficient on this variable is never statistically or economically significant.

Second, to further rule out reverse causality we add the growth of a bank's branches to the set of control variables. This approach allows us to rule out the notion that banks with strong loan demand open or purchase new branches to fund loan growth. Specifically, if banks expand their branch network in anticipation of a boom, this additional explanatory variable should take power away from our core variable of interest, exposure to the fracking boom. We find a positive correlation between branch growth and mortgage growth, but adding this variable has almost no impact on the coefficient of interest.

Third, we evaluate the possibility that our results are driven by banks with branches in close proximity to shale-boom counties growing faster than other banks in the same county. If some banks' lending grows faster due to demand spillovers from neighboring boom counties, then our results could be driven by both supply- and demand-side shocks. It is also possible that some of the additional deposits in shale-boom counties come from nearby counties due to migration into the booming areas. We test these hypotheses by excluding bank-county-year observations from counties sharing a border with a boom county. The results are similar to those reported in our baseline models in terms of both statistical and economic magnitude and further support the notion that the effects we document are supply-side driven.

Fourth, the summary statistics presented in Table I indicate that the exposed banks tend to be larger than those never exposed to shale booms. The disparity occurs because large banks, by the very fact that they are large, have a greater likelihood of having at least some exposure to counties with shale booms. Large banks, however, also have wide access to the capital markets and, during the time of crisis, government financial support, and hence might grow their lending quicker than the rest of the banking sector. To evaluate this conjecture, we estimate our model without very large banks, defined as those in the top decile of the asset size distribution. The coefficient on *Share of Branches in Boom Counties* increases slightly in magnitude and statistical significance when we impose this filter.

Fifth, we estimate our model after dropping bank-county-years where the mortgage growth rate is based on fewer than 15 loans during the prior year. This filter drops observations likely to have substantial noise in the dependent variable. Again, the results are stronger than before, in terms of both

magnitudes and statistical significance. This indicates that our results cannot be attributed to noise in measuring changes in banks' origination decisions.

Sixth, we add bank*county fixed effects. Adding these effects removes the possibility that some banks may always grow faster than others within the same county. For example, some banks may simply advertise more in specific areas or have more branches in better locations, leading to persistently higher rates of mortgage growth. In fact, adding bank*county effects increases the magnitude and statistical significance of our results.

Seventh, we test whether inflows affect mortgage growth most in those markets with the highest unserved credit demand. To measure unserved credit demand, we follow Mian and Sufi (2009), who argue that the advent of subprime credit had its greatest impact on neighborhoods with unmet demand for mortgage credit, based on the mean mortgage approval rate in the area at the beginning of their sample. We apply their strategy to our setting by interacting our measure of shale exposures with the average mortgage approval rate (based on HMDA data) from all mortgage applications made during the prior bank-county-year. Consistent with efficient capital flows across regions, shale-boom windfalls spur lending most in areas with low mortgage approval rates in the prior year, which we interpret as a proxy for unsatisfied demand for mortgage credit.

G. Aggregate Effects

In our core set of tests, we isolate the supply effect of bank liquidity windfalls by comparing exposed and nonexposed banks' lending in the *same* county-year. County-year effects fully absorb credit demand, but they also absorb any potential aggregate effect of credit supply. Our explanation for the results: banks exposed to shale booms make new loans, thus raising aggregate credit supply. It is also possible, however, that banks with access to positive liquidity inflows simply outcompete banks not exposed to booms. In this case, credit would be reallocated between banks in a county but aggregate credit supply would not rise.

In our last set of tests, we attempt to discriminate between these two hypotheses by evaluating the growth in overall originations at the county level. Identification at this level of aggregation is less compelling because we have no way to fully absorb demand. We go as far as possible to absorb demand. First, we include time-invariant county effects (as opposed to time-varying county effects, as in equation (2)) along with an overall time effect. Second, we report the model with state-specific time effects. Thus, we estimate the following regressions:

$$\begin{aligned} \textit{Mortgage Growth}_{j,t} = \alpha_j + \alpha_t + \beta\ \textit{County Boom Exposure}_{j,t} \\ + \textit{County Controls}_{j,t} + \varepsilon_{i,j,t}. \end{aligned} \tag{3a}$$

Table VII

The Effect of Liquidity on County-Level Aggregate Mortgage Lending

This table reports regressions of the percentage change in county-year mortgage originations (excluding mortgages for refinancing) on each county's aggregate connection via bank branch network to the shale-boom shock. The sample includes counties that are not exposed directly to the shale booms. Standard errors are clustered by county. t-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Counties in All States		Counties in Seven States with Shale Booms	
	(1)	(2)	(3)	(4)
Average Share of Branches in Boom Counties				
Weighted by number of branches	0.168***	0.0907**	0.0697*	0.0900**
	(5.00)	(2.18)	(1.83)	(2.17)
Year fixed effects	Yes	–	Yes	–
County fixed effects	Yes	Yes	Yes	Yes
State*year fixed effects	–	Yes	–	Yes
Local time-varying macro controls	Yes	Yes	Yes	Yes
Observations	33,889	33,889	6,947	6,947
R^2	35.80%	40.20%	24.50%	26.30%

$$Mortgage\ Growth_{j,t} = \alpha_{j+}\alpha_{t,k} + \beta\ County\ Boom\ Exposure_{j,t} + County\ Controls_{j,t} + \varepsilon_{i,j,t}. \tag{3b}$$

The dependent variable in (3a) and (3b)—$Mortgage\ Growth_{j,t}$—omits mortgages for refinancing because, as we documented above, shale-boom liquidity windfalls have no effect on growth in mortgage refinancings. $County\ Boom\ Exposure_{j,t}$ equals the average measure of shale boom exposure across all banks operating in the county-year, weighted by each bank's share of branches in the county. Beyond the fixed effects, we also include a set of time-varying county level controls for economic conditions (contemporaneous employment growth, payroll growth, and population growth). As in our earlier tests, we include only connected counties and omit those counties that experienced shale booms between 2003 and 2010.

The results (Table VII) suggest that aggregate credit supply increases with average bank exposure to the shale booms. This conclusion holds using either the seven states that experienced shale booms as well as all states. The coefficient on exposure suggests that a one SD increase in county-level exposure (= 0.07) leads to an increase in mortgage growth of about 0.6% per year (using the coefficients from column (2) or (4), which are quite similar). These results suggest that the increase in lending among bank-counties exposed to shale booms translates into higher mortgage origination, rather than a mere redistribution of mortgages across lenders.

IV. Conclusions

Shale-boom discoveries provide large and unexpected liquidity windfalls at banks with branches nearby as mineral rights owners pay back old debt and deposit large amounts of their new wealth into local banks. Banks export these liquidity windfalls into outlying (nonboom) markets and increase mortgage lending, but *only* when such banks have branches in *both* markets. Banks experiencing inflows do not export liquidity and lend more in areas where they have no branch presence because, we argue, without a branch presence banks have no advantage over securitization markets. Combining these results, we conclude that, despite the advent of securitization, bank branch networks remain important in fully integrating the mortgage market. By extension, credit markets in which securitization and arm's length finance have had less of an impact, such as small business lending, rely even more heavily on branch networks to move capital across markets.

Our results provide evidence that bank branching fosters financial integration by allowing savings collected in one locality (shale-boom counties) to finance investments in another (nonboom counties). The result is important for two reasons. First, it demonstrates the limits to arm's length financing technologies such as securitization in integrating financial markets. For credit markets requiring lenders to locate near borrowers to adequately understand and monitor risk, securitization is not a viable financing mechanism. Second, it suggests that, by allowing capital to flow more easily across local markets, deregulation of bank branching fostered a denser branch network that improved capital mobility and investment allocation efficiency.

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Supporting Information

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Appendix S1: Internet Appendix.